

KEERTHANA B V

AI Solutions Engineer | Voice AI, Prompt Engineering & LLM Agents

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#) (live chat & voice bot demos) | [Research \(Google Scholar\)](#)

PROFESSIONAL SUMMARY

Voice AI and LLM developer who has designed, prompted, and shipped production conversational agents end to end: a real-time multilingual (English/Tanglish) Vapi voice agent for live order intake, and a deployed support RAG agent with 94.4% faithfulness (RAGAS) and a 100% prompt-injection block rate from systematic red-teaming. Combines deep hands-on prompting with client-facing experience across 5+ customers.

CORE SKILLS

- **Prompt Engineering:** system prompt design, dialogue flow design, multi-turn context handling, structured outputs (JSON Schema), slot-filling, hallucination mitigation, red-teaming & prompt-injection defense
- **Voice AI:** Vapi AI (streaming STT/TTS), Twilio, conversation/call-flow design, vernacular & code-switched speech (English/Tanglish), latency-aware turn-taking
- **LLM Tooling:** OpenAI GPT-4o-mini, Llama-3.1 (Groq), LangChain, RAG, RAGAS evaluation, FAISS/BM25 retrieval, context windows, temperature & sampling
- **Automation & Integration:** Make.com, webhooks, Google Sheets pipelines, REST APIs, error-handling routes, no-code/low-code workflows
- **Engineering & Customer-Facing:** Python, JavaScript, SQL, FastAPI, React.js, Git; requirement gathering, client demos, UAT, documentation

KEY PROJECTS

Real-Time Conversational Voice AI Agent

Vapi AI, GPT-4o-mini, Twilio, Make.com

- Designed and built a real-time, multilingual (English/Tanglish) voice agent using Vapi's streaming STT/TTS pipeline and GPT-4o-mini, handling live customer order intake and L1 support with natural, low-latency turn-taking.
- Wrote and iterated system prompts and dialogue flows with prompt-level guardrails to keep multi-turn conversations on-task, extracting structured entities (name, order, phone) via JSON Schema with reliable slot-filling across accented, code-switched speech.
- Built a fault-tolerant automation pipeline (Make.com webhooks + Twilio SMS payment links) with custom error-handling routes so downstream API failures never interrupted a live call — a core requirement for always-on voice systems.

AI Customer Support Conversational Agent ([live demo](#))

FastAPI, LangChain, Llama-3.1, React

- Architected and deployed a full-stack e-commerce support RAG agent streaming real-time responses via SSE at sub-second latency; live on Render + Vercel.
- Red-teamed the agent with adversarial inputs and built double-layer prompt-injection defenses (keyword screening + LLM classification), achieving a 100% block rate on the attack test suite.
- Validated answer quality with a custom RAGAS evaluation runner — 94.4% faithfulness and 85.5% context precision — plus an 8-case automated suite covering intent routing, escalation, and injection attempts.
- Engineered stateful intent routing with multi-turn sentiment tracking that auto-escalates frustrated customers to ticketed human handoff.

AI Legal Document Intelligence Agent (Best Paper, NCRIE 2025)

Legal-BERT, LoRA, PyTorch, Hugging Face

- Fine-tuned a domain-specific BERT model with LoRA to 84.9% F1 on the 510-contract CUAD benchmark in ~3 hours on a single T4 GPU; published with Best Paper Distinction.
- Applied evidence-driven evaluation (precision, recall, confusion-matrix analysis) — the same measurement mindset used to judge where agents fail and fix them.

WORK EXPERIENCE

Full Stack Developer | ASPL Tech Solutions Pvt. Ltd., Bengaluru

Oct 2025 – Mar 2026

- Collaborated directly with 5+ clients to gather requirements, run project updates, and align deliverables with business needs hands-on experience translating customer goals into working software.
- Developed client-facing web applications (React.js, Node.js) for healthcare and retail; built an automated HRMS onboarding module and implemented RBAC + JWT authentication.

MERN Stack Developer Intern | Dyashin Technosoft Pvt. Ltd., Bengaluru

Nov 2024 – Jan 2025

- Built a full-stack MERN e-commerce platform (JWT auth, 15+ documented REST endpoints); resolved 15+ critical UAT bugs, cutting post-launch defects by 40%.

EDUCATION

MCA — RV Institute of Technology and Management, Bengaluru | CGPA 8.2/10 | Aug 2025

BCA — Community Institute of Commerce and Management | CGPA 8.4/10 | 2023

Languages: English (professional) • Kannada • Telugu